

The Agentic AI Bootcamp

Weekly Build: Multi-Agent Support System

By Dr. Ryan Ahmed & Kukesh Kodess

The objective:

Design a cost-efficient AI support system for CloudSync, a B2B SaaS company handling 15,000 tickets/month using OpenAI Agent Builder. The company receives various tickets that vary in their complexity:

- 60% are “Simple” tickets such as FAQ lookup
- 25% are “Medium”, such as Billing and account issues
- 15% are “Complex”, like technical failures and cancellation risk

Using the same Large Language model (LLM) for all tickets is not recommended, as the company can overspend if all tickets use an expensive model (e.g., GPT-5.2) or fail on critical tickets if all are directed to a cheaper GPT-5-nano model. Here is what it costs if you use one model for everything:

Your Task:

Build a routed multi-agent system in OpenAI Agent Builder that handles these tickets intelligently. Follow these three steps.

1. Identify Ticket Groups

Read the 5 sample tickets in the appendix and define logical ticket categories, identify high-stakes cases, and briefly justify your grouping.

2. Build a Classifier Agent

Build an agent that reads a ticket and decides which group it belongs to.

- This agent does not answer the ticket; it only classifies it.
- Use structured output (JSON) so the next step knows exactly which group was picked.
- Think about what happens when the classifier is not confident. Where should uncertain tickets go?

3. Build a Handler Agent for each category

Each group gets its own agent. Each agent gets the relevant data below in its system prompt.

Hints:

- The FAQ data below goes into the FAQ handler's system prompt.
- The Account & Billing Notes go into the billing handler's system prompt.
- The Account Policies & Service Level Agreements (SLAs) go into the escalation handler's system prompt.
- Not every agent needs the same model. Match the model to the job.

All agent prompts must follow the Context, Instruction, Input, and Output indicator format.

Available models:

Model	Cost	Best For
GPT-5-nano	\$0.001 / call	Simple tasks, classification, pattern recognition
GPT-5-mini	\$0.01 / call	Medium complexity, can use tools
GPT-5.2	\$0.05 / call	Complex reasoning, nuanced responses

Available tools: Web Search. You decide which agents need it.

Deliverables

1. Submit a working multi-agent system in OpenAI Agent Builder. A screenshot should be sufficient.
2. Screenshots of each sample ticket being handled
3. 1-page architecture explanation including: (1) Model choices, (2) Routing logic, (3) Why this beats a single-model system

APPENDIX

Data: FAQ Knowledge Base

Use this as reference data in your system "agent system prompt". Copy and paste where needed.

CloudSync FAQ v1

Q: I forgot my password and reset email is not arriving.

A: Ask the user to check spam/promotions, verify account email, and request another reset link after 2 minutes. If still missing, whitelist no-reply@cloudsync.com and contact support with email + timestamp.

Q: How do I add a new team member?

A: Go to Settings -> Team -> Invite Member, enter email, choose role (Admin, Editor, Viewer), click Send Invite. Invite expires in 7 days.

Q: How do I change my subscription plan?

A: Go to Settings -> Billing -> Change Plan. Upgrades take effect immediately. Downgrades take effect at the next billing cycle.

Q: Where can I find invoices?

A: Go to Billing -> Invoices. Billing Admin can download PDF invoices.

Q: What is the difference between Pro and Enterprise?

A: Enterprise includes SSO/SAML, higher API limits, audit log export, priority SLA, and advanced retention controls.

Data: Account & Billing Notes

Reference data for account and billing tickets.

CloudSync Account & Billing Notes v1

Billing cycle:

- All plans bill on the 1st of each month
- Mid-cycle upgrades are prorated immediately
- Mid-cycle downgrades take effect next billing cycle
- Refunds for double charges: process within 3-5 business days

Subscription changes:

- Pro to Enterprise: contact sales for custom pricing

- Enterprise to Pro: 30-day notice required per contract
- Cancelled accounts retain read-only access for 30 days

Common sync issues:

- "Data not syncing" is usually a permission issue, not a bug
- Workspace owner must grant sync access to each member
- Sync runs every 15 minutes, not real-time
- If sync stalls, disconnect and reconnect the integration

Team permissions:

- Admin: full access, can manage billing and team
- Editor: can create and edit, cannot manage billing
- Viewer: read-only access to shared workspaces

Data: Account Policies & SLAs

Reference data for high-impact customer situations.

CloudSync Account Policies v1

SLA by plan:

- Enterprise: 99.9% uptime guarantee, 4-hour response time
- Pro: 24-hour response time, no uptime guarantee
- Free: community support only, no SLA

Compensation for service issues:

- Enterprise: 1 day account credit per hour of downtime
- Pro: pro-rated credit for outages over 24 hours
- Credits must be requested within 30 days of the incident

Account management:

- All Enterprise accounts have a dedicated account manager
- Account manager email: am-team@cloudsync.com
- Pro accounts: no dedicated manager, general support only

Data export:

- Full data export available within 48 hours of request
- Request via Settings -> Data -> Export All
- Exported data delivered as CSV files via secure download link

Cancellation policy:

- Enterprise: 30-day written notice required per contract
- Pro: cancel anytime, access continues until end of billing cycle
- All plans: pro-rated refund available within first 30 days only

Sample Tickets

Your system must handle all 5 of these tickets. Use them to test your build.

Ticket 1

Subject: Can't reset my password

Hi, I forgot my password and the reset email isn't coming through. How do I get back into my account?

Ticket 2

Subject: Where can I find my invoices?

I need to download our last 3 months of invoices for our finance team. Where do I find them?

Ticket 3

Subject: Charged twice this month

I just checked my credit card statement, and I see two charges from CloudSync for this month. We are on the Pro plan. Can you fix this?

Ticket 4

Subject: Data not syncing for our team

Three people on our team say their data stopped syncing yesterday. I am the workspace owner. We have tried logging out and back in but it is still not working.

Ticket 5

Subject: Considering cancellation after repeated issues

This is the third time this quarter we have had problems with our account. Last month we were charged incorrectly, two weeks ago half our team lost access, and now our data is not syncing again. Leadership is reviewing alternatives. We need someone to take ownership of this.